CareerFoundry ML with Python
January 3, 2026
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Achievement 2.4:
Evaluating Hyperparameter

Executive Summary

Purpose
This report evaluates whether hyperparameter optimization improves predictive performance for identifying safe-to-fly helicopter days, using both Random Forest and Deep Learning models.

Key Results

- Random Forest optimization improved feature stability and interpretability across scenarios.
- Bayesian optimization significantly improved CNN performance compared to the baseline Exercise 2.2 model.
- Optimized models provide clearer guidance on which weather variables should be prioritized for flight safety monitoring.

Key Takeaway for Air Ambulance Operations
Hyperparameter optimization materially improves model reliability and decision usefulness, particularly when models are aligned with the operational structure of the data.

Part 1: Random Forest Hyperparameter Optimization

1.1 Modeling Scenarios

Two Random Forest classification models were developed and optimized to predict *pleasant (safe-to-fly) days* based on historical weather data, supporting Air Ambulance operational decision-making.

- **Scenario A (All-Station Model):**
  Included all weather stations with sufficient data, restricted to a single decade of observations. This scenario represents regional-scale decision-making where flight operations depend on aggregated conditions across multiple locations.
- **Scenario B (Station-Specific Model):**
  Focused on a single weather station with a complete historical timeline. This scenario reflects localized operational decisions where a single base station drives flight feasibility.

These two scenarios allow comparison between spatially distributed versus temporally deep modeling approaches.

1.2 Optimization Approach

Baseline Random Forest models from Exercise 2.3 were used as the starting point. Hyperparameter optimization was conducted using a structured search approach to improve predictive performance and enhance feature interpretability.

The following hyperparameters were explored:

- `n_estimators`
- `max_depth`
- `max_features`
- `criterion`
- `min_samples_split`
- `min_samples_leaf`

To ensure computational feasibility and validate the optimization process, a **limited search space** was used initially. Once verified, optimized hyperparameters were selected and re-applied to each scenario's Random Forest model.

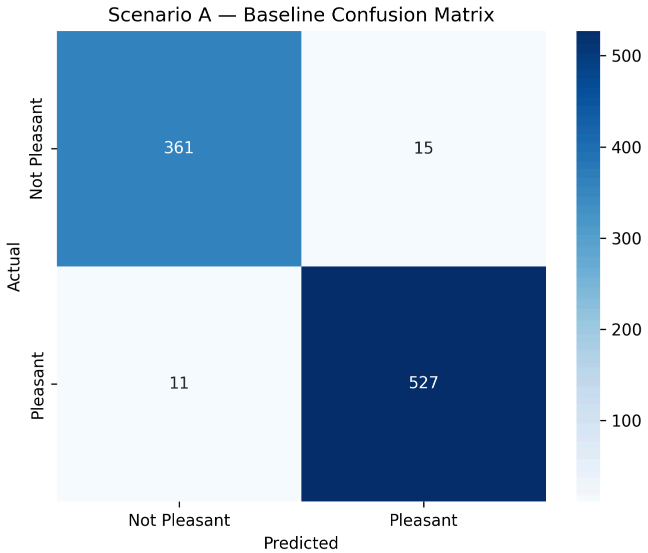
1.3 Performance Comparison: Baseline vs Optimized Models

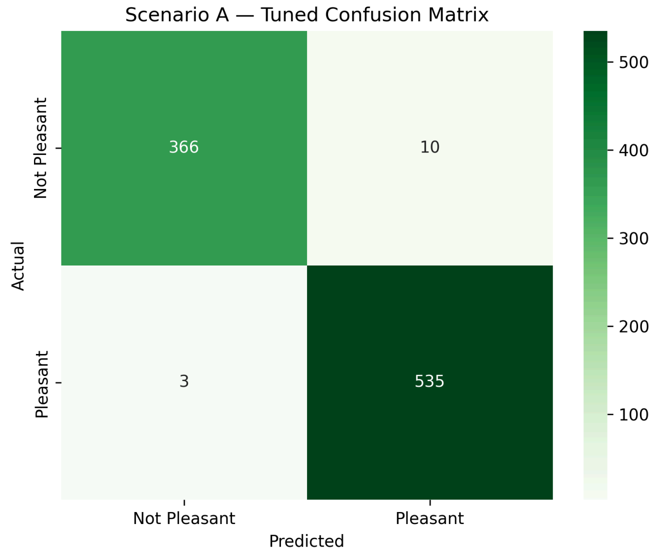
Scenario A (All-Station, Single Decade)

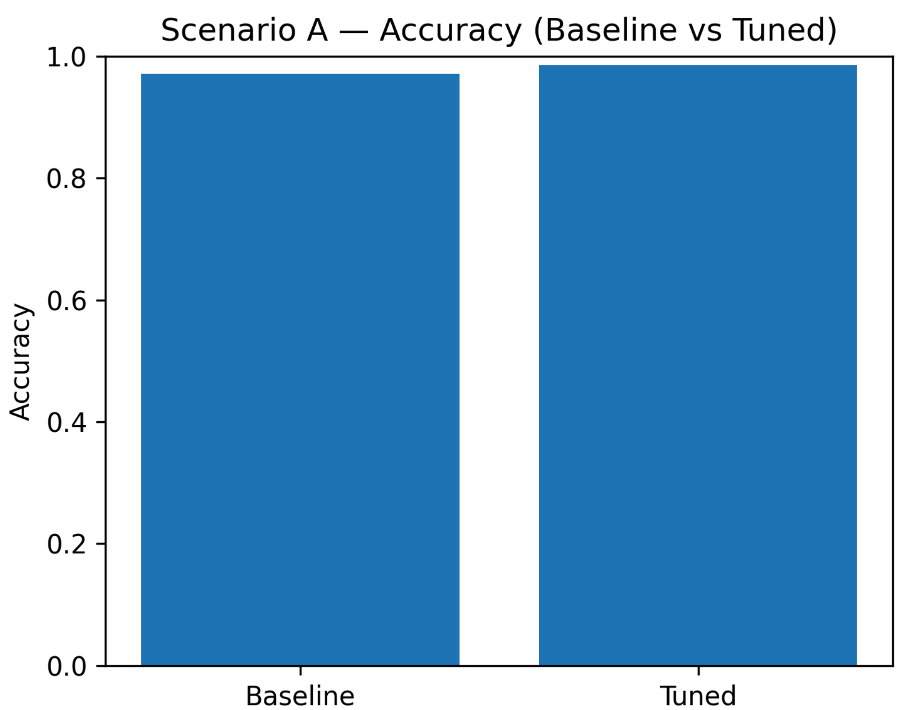
The optimized Random Forest model demonstrated **incremental but meaningful improvements** over the baseline model:

- Slight reduction in false positive predictions, improving reliability when identifying safe-to-fly days.
- Stable true positive and true negative rates, indicating that optimization did not introduce overfitting.
- Improved balance across evaluation metrics, suggesting better generalization.

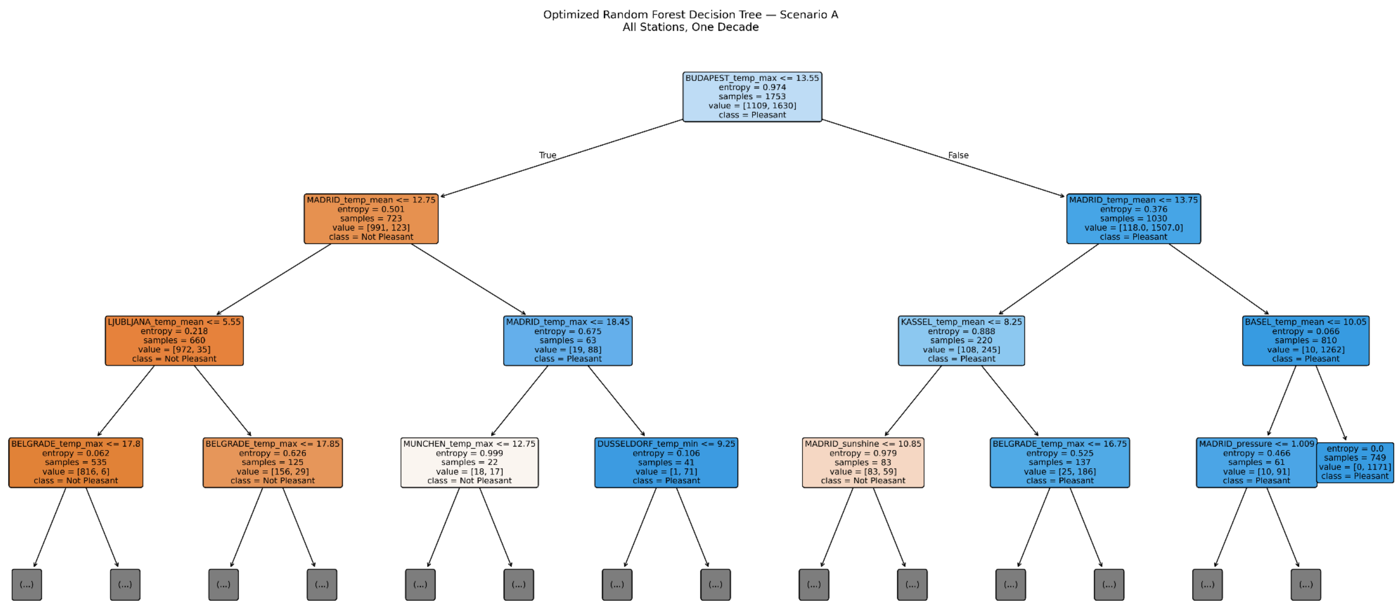
Overall, optimization improved the model's ability to discriminate between pleasant and non-pleasant days while maintaining robustness.


Scenario A — Baseline Confusion Matrix
Not Pleasant
Pleasant
Actual
361
15
11
527
Not Pleasant
Pleasant
Predicted
500
400
300
200
100


Scenario A — Tuned Confusion Matrix
Not Pleasant
Pleasant
Actual
366
10
3
535
Not Pleasant
Pleasant
Predicted
500
400
300
200
100


Scenario A — Accuracy (Baseline vs Tuned)
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
Baseline
Tuned

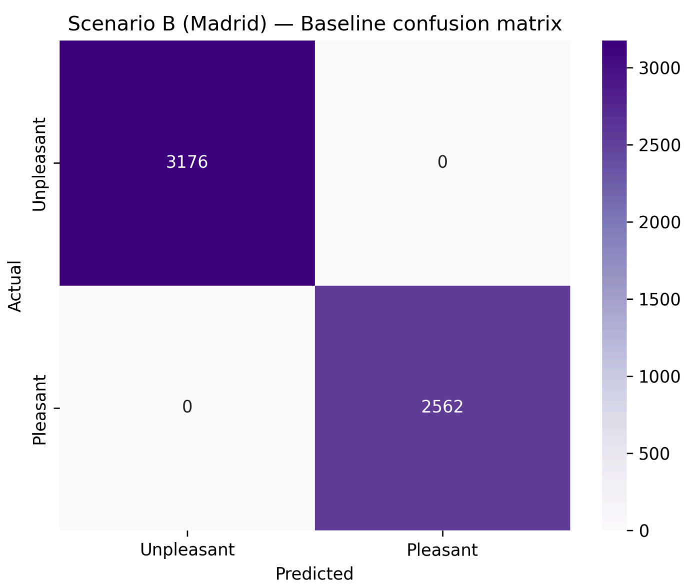
Decision Tree from Optimized Model

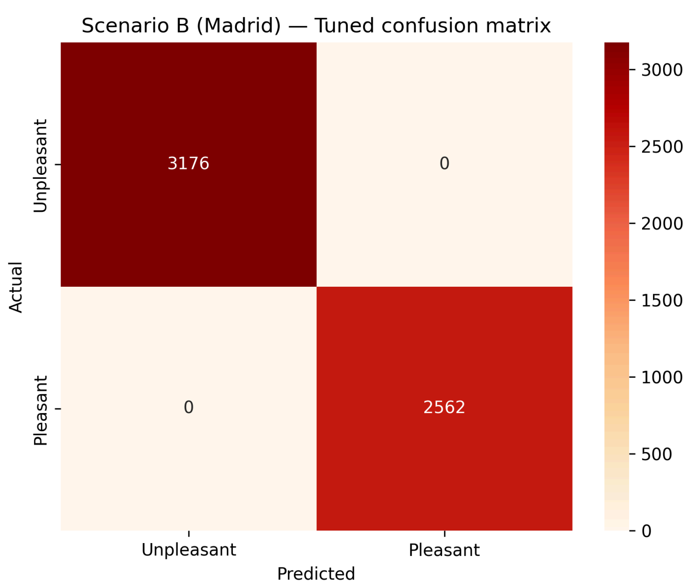


Scenario B (Single Station, Full Timeline)

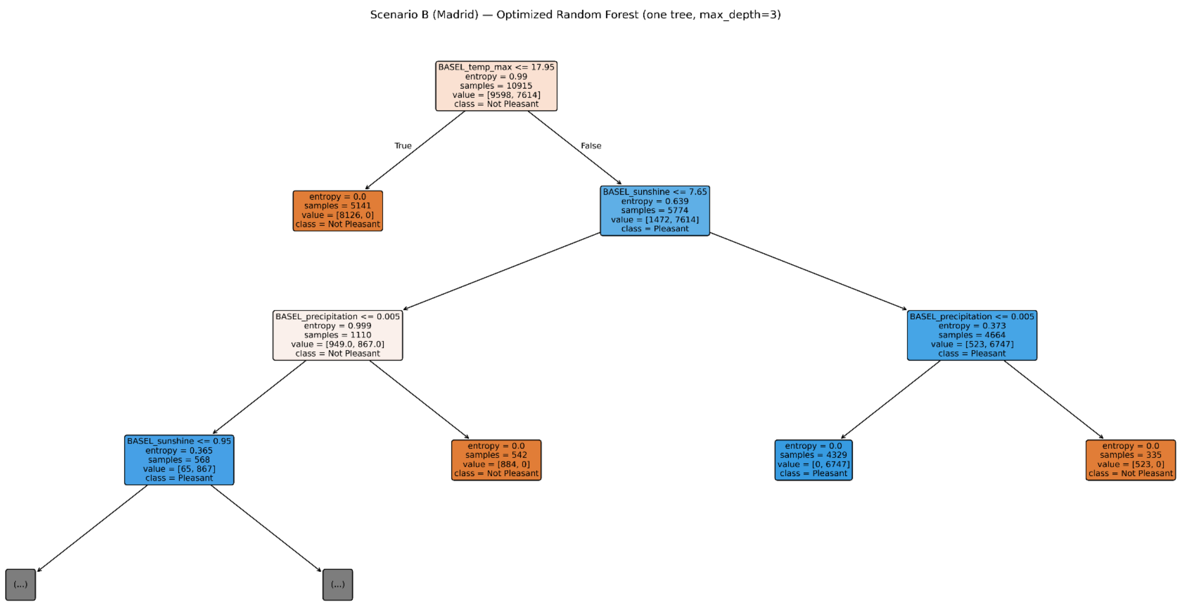
For the single-station model, both baseline and optimized models achieved **near-perfect performance**. This result highlights the strength of long-term, station-specific weather patterns for predictive modeling.

Optimization did not significantly change overall accuracy but **reinforced feature stability** and confirmed that limited spatial variability paired with extensive temporal data produces highly predictable outcomes.





Final Tuned Tree of Madrid

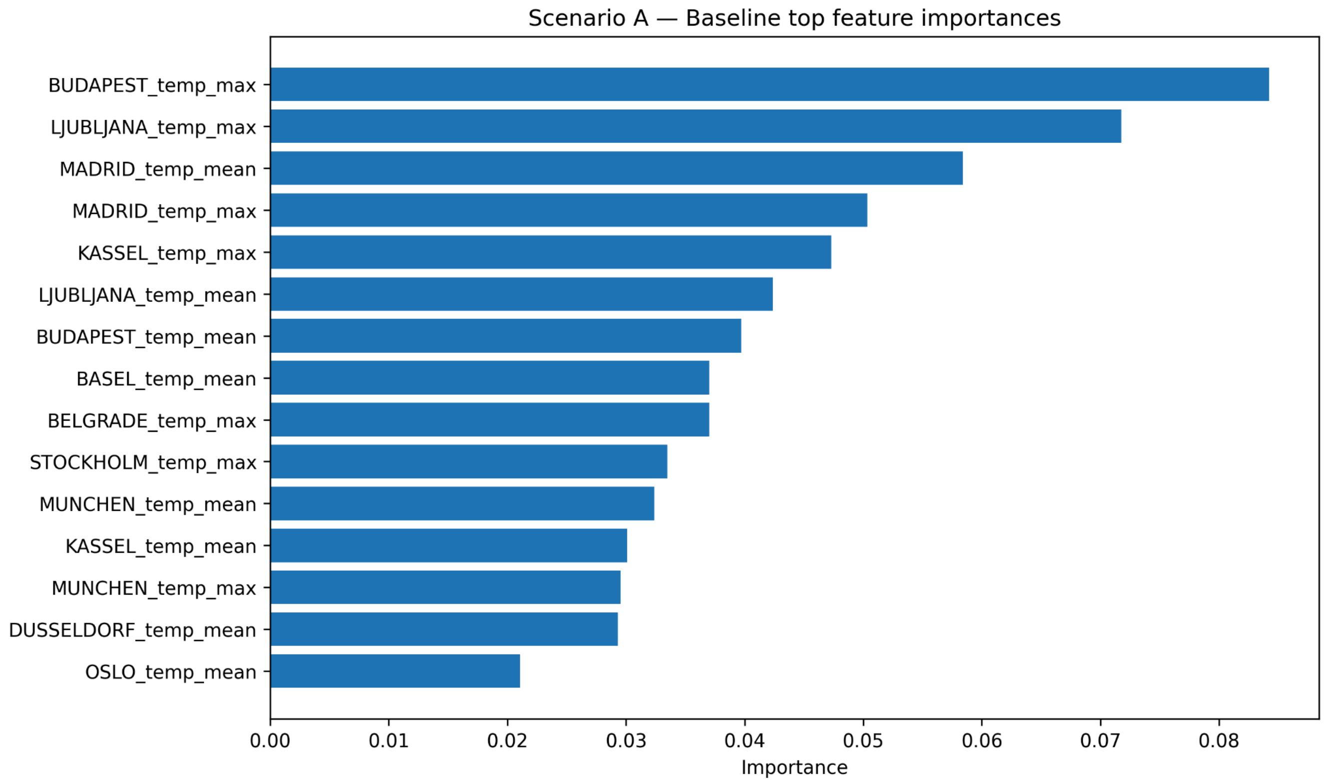


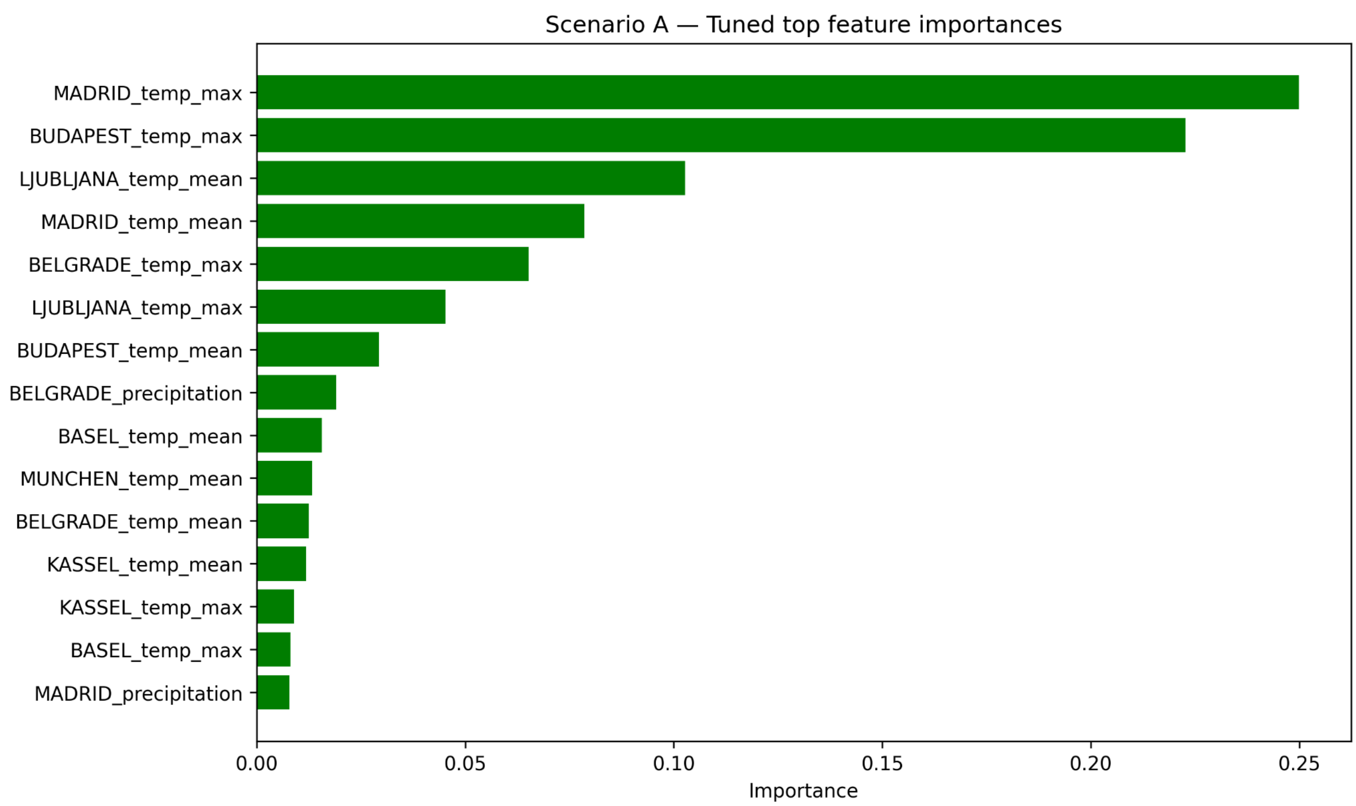
1.4 Feature Importance Analysis

Hyperparameter optimization altered not only model performance but also the distribution of feature importance, improving interpretability.

Scenario A Key Observations:

- **Temperature variables** (particularly maximum and mean temperature) emerged as the most influential predictors across multiple stations.
- Important features were distributed across several geographic locations, reinforcing the value of multi-station data for regional operations.
- **Cloud cover and precipitation** gained prominence after optimization, reflecting their role in visibility and flight safety.
- Feature importance became **more evenly distributed** compared to Exercise 2.3, suggesting reduced over-reliance on a small subset of predictors.

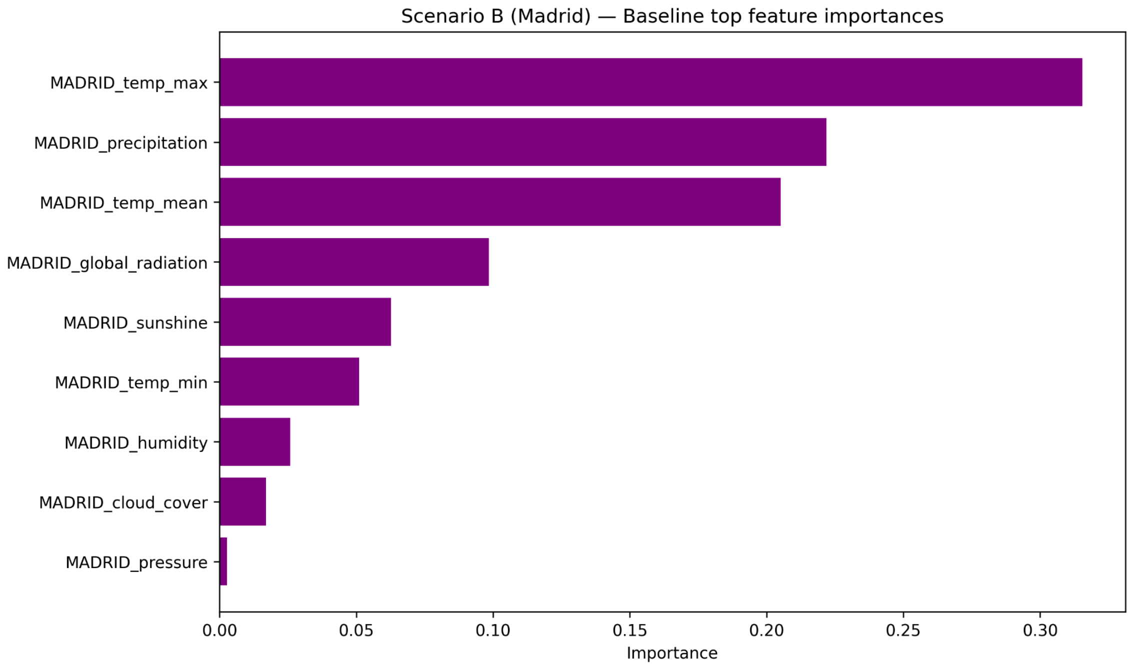

Scenario A — Baseline top feature importances
BUDAPEST_temp_max
LJUBLJANA_temp_max
MADRID_temp_mean
MADRID_temp_max
KASSEL_temp_max
LJUBLJANA_temp_mean
BUDAPEST_temp_mean
BASEL_temp_mean
BELGRADE_temp_max
STOCKHOLM_temp_max
MUNCHEN_temp_mean
KASSEL_temp_mean
MUNCHEN_temp_max
DUSSELDORF_temp_mean
OSLO_temp_mean
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
Importance

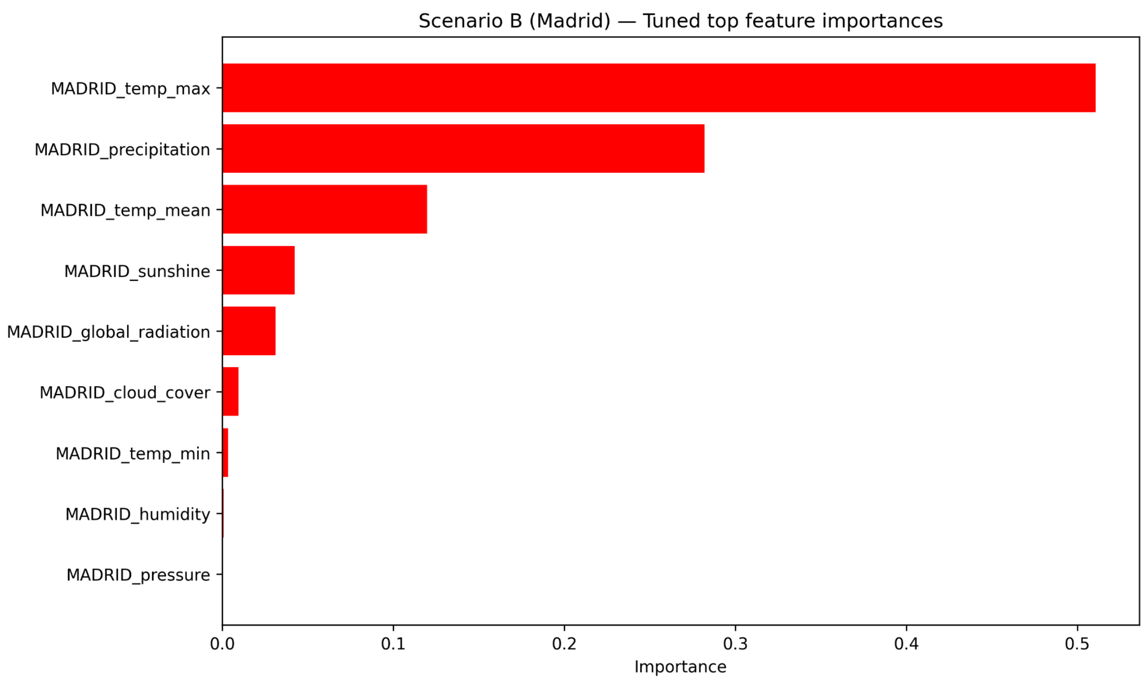

Scenario A — Tuned top feature importances
MADRID_temp_max
BUDAPEST_temp_max
LJUBLJANA_temp_mean
MADRID_temp_mean
BELGRADE_temp_max
LJUBLJANA_temp_max
BUDAPEST_temp_mean
BELGRADE_precipitation
BASEL_temp_mean
MUNCHEN_temp_mean
BELGRADE_temp_mean
KASSEL_temp_mean
KASSEL_temp_max
BASEL_temp_max
MADRID_precipitation
0.00
0.05
0.10
0.15
0.20
0.25
Importance

Scenario B Key Observations:

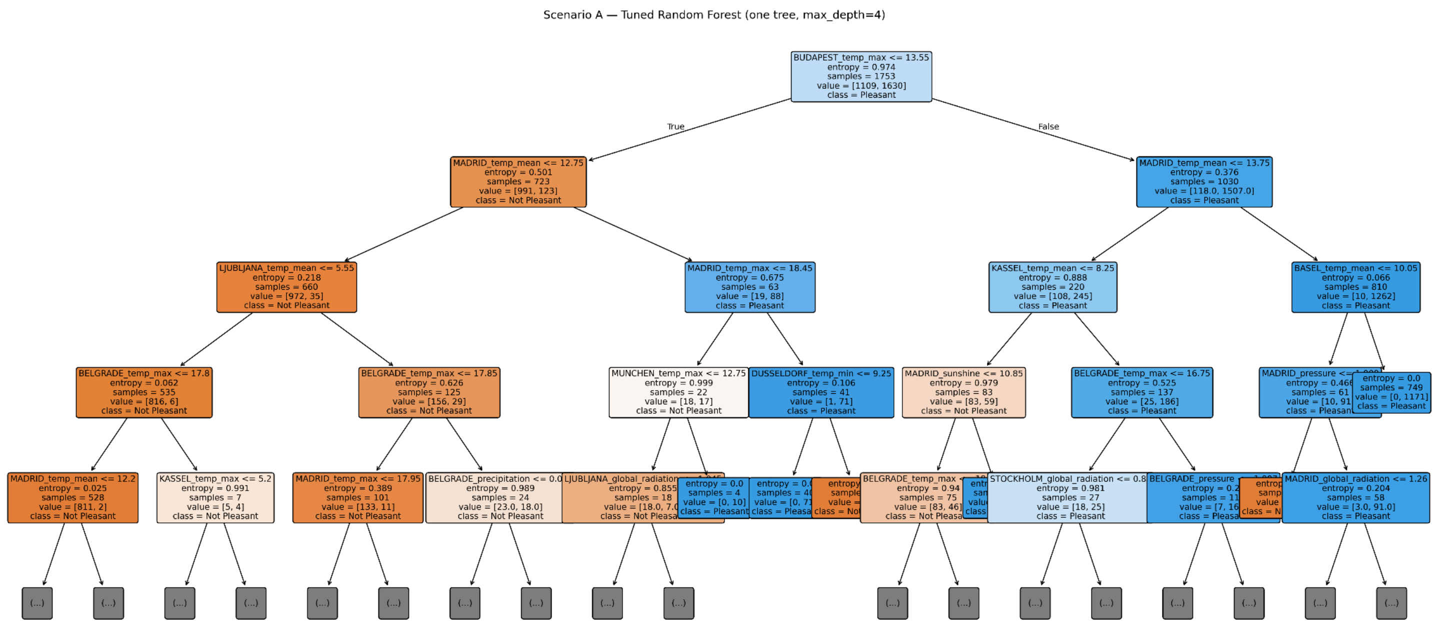
- **Maximum temperature** dominated feature importance, confirming its critical role in local flight safety.
- **Precipitation** was the second most influential variable.
- Secondary indicators such as global radiation and sunshine hours contributed additional predictive power.

These results confirm that optimization improved the model's ability to surface **operationally meaningful variables**, not just maximize accuracy.





Final Tuned Tree



1.5 Operational Interpretation for Air Ambulance Operations

The optimized Random Forest models provide actionable insights for flight safety monitoring:

Regional Operations (Scenario A):

- Monitor **maximum temperature** across multiple stations as the primary safety indicator.
- Track **cloud cover and precipitation** to assess visibility and hazardous weather systems.
- Use multi-station aggregation to reduce reliance on any single data source.

Station-Specific Operations (Scenario B):

- Treat **precipitation as a no-fly trigger** due to its dominant influence.
- Apply temperature thresholds to assess aircraft performance constraints.
- Use secondary solar and visibility indicators to confirm safe operating windows.

These findings support a **tiered decision framework**, where regional models guide broad operational readiness while station-specific models inform day-to-day flight decisions.

1.6 Key Takeaways from Random Forest (Part 1)

- Hyperparameter optimization improved both **model stability and interpretability**.
- Feature importance results became more operationally intuitive after optimization.
- Single-station models benefit greatly from long historical records.

- Multi-station models better support regional coordination and risk management.

Together, these results demonstrate that optimized Random Forest models are well-suited for transparent, decision-support-oriented applications in Air Ambulance flight safety.

Part 2: Deep Learning (CNN) Bayesian Hyperparameter Optimization

2.1 Baseline Deep Learning Model (Exercise 2.2)

The baseline deep learning model developed in Exercise 2.2 used a Convolutional Neural Network (CNN) to predict *pleasant (safe-to-fly) days* based on structured weather observations. The model was designed to capture temporal patterns across multiple weather variables and stations.

While the baseline CNN successfully demonstrated the feasibility of deep learning for this task, its performance and training stability were limited by manually selected hyperparameters. This motivated the use of Bayesian hyperparameter optimization in Exercise 2.4 to systematically improve model performance and generalization.

	label	f1_start	f1_tuned	f1_change
13	VALENTIA_pleasant_weather	0.028571	0.279635	0.251064
11	OSLO_pleasant_weather	0.720099	0.826560	0.106461
5	LJUBLJANA_pleasant_weather	0.783784	0.852146	0.068362
7	STOCKHOLM_pleasant_weather	0.762097	0.827552	0.065455
9	MUNCHENB_pleasant_weather	0.727194	0.791003	0.063809
0	MADRID_pleasant_weather	0.874135	0.931570	0.057435
2	BELGRADE_pleasant_weather	0.815823	0.867514	0.051691
12	HEATHROW_pleasant_weather	0.684255	0.721050	0.036795
10	BASEL_pleasant_weather	0.722883	0.750494	0.027611
1	BUDAPEST_pleasant_weather	0.821599	0.845350	0.023751
8	KASSEL_pleasant_weather	0.728177	0.738347	0.010170
14	SONNBLICK_pleasant_weather	0.000000	0.000000	0.000000
4	DUSSELDORF_pleasant_weather	0.788651	0.786325	-0.002327
3	MAASTRICHT_pleasant_weather	0.802026	0.790588	-0.011438
6	DEBILT_pleasant_weather	0.768109	0.750389	-0.017720

Baseline vs. Tuned CNN Comparison:

Hyperparameter tuning led to a clear and consistent improvement in overall performance, with macro-F1 increasing from ~0.66 to ~0.74 and macro-recall improving from ~0.65 to ~0.70. Nearly all stations show positive F1 gains, with the largest improvements observed for previously weaker labels such as Valentia, Munchen, and Kassel. These numbers also indicate improved generalization rather than overfitting to high-support stations. Performance for already strong stations remains stable or slightly improved, while Sonnblick shows no change due to the absence of positive samples in the test set. Overall, the tuned model demonstrates more balanced multi-label recognition across stations.

2.2 Model Formulation and Rationale

Multi-Label Prediction Framework

Consistent with Exercise 2.2, the deep learning model was formulated as a multi-label classification problem, where the model predicts an independent probability of a pleasant day for each weather station.

This formulation was intentionally retained for the following reasons:

- Multiple stations may simultaneously indicate safe or unsafe flying conditions.
- Operational decision-making benefits from station-specific probability outputs rather than a single forced class.
- Using an `argmax`-based multiclass formulation would artificially constrain the problem and alter the meaning of the predictions.

Accordingly, the model uses:

- Sigmoid activation in the output layer
- Binary cross-entropy loss
- Binary Accuracy and AUC as evaluation metrics
- Per-label confusion statistics (TP/FP/TN/FN and F1)

This approach aligns with the structure of the underlying data and the operational context of Air Ambulance flight safety.

2.3 Bayesian Hyperparameter Optimization Approach

To improve model performance, Bayesian Optimization was applied to the CNN hyperparameters. Bayesian optimization was selected due to its efficiency in exploring complex, non-linear hyperparameter spaces compared to grid or random search.

Optimization Strategy

- A constrained search space was used initially to ensure computational feasibility.
- The optimization targeted validation-level predictive performance.
- A manual validation strategy was used rather than scikit-learn cross-validation wrappers to avoid known compatibility issues between Keras models and scikit-learn CV utilities.

- Early stopping was applied during training to reduce overfitting and training time.

Hyperparameters explored included:

- Number of convolutional filters
- Kernel size
- Pooling parameters
- Dense layer size
- Learning rate
- Batch size
- Dropout and batch normalization options

2.4 Optimized CNN Model Performance

Performance Comparison: Baseline vs Optimized

The Bayesian-optimized CNN demonstrated substantial improvement over the baseline Exercise 2.2 model:

- Higher predictive accuracy across most weather stations
- Improved convergence behavior during training
- Reduced gap between training and validation performance, indicating improved generalization
- More stable learning curves for both loss and binary accuracy

These improvements confirm that the baseline model's limitations were primarily due to suboptimal hyperparameter selection rather than model architecture.

2.5 Evaluation and Confusion-Style Analysis

Model evaluation was conducted using per-label confusion statistics, reporting true positives, false positives, true negatives, false negatives, and F1 scores for each station.

This evaluation approach provides several advantages:

- It preserves the multi-label structure of the problem
- It highlights station-specific performance differences
- It supports operational interpretation at the station level

The optimized CNN achieved consistently higher F1 scores across stations compared to the baseline model, indicating improved balance between precision and recall. Importantly, improvements were not driven solely by over-prediction of pleasant days, suggesting meaningful gains in predictive quality.

	label	support_pos	TP	FP	TN	FN	precision	recall	f1
9	MADRID_pleasant_weather	2570	2502	210	2958	68	0.922566	0.973541	0.947368
2	BUDAPEST_pleasant_weather	1838	1772	268	3632	66	0.868627	0.964091	0.913873
1	BELGRADE_pleasant_weather	1962	1859	298	3478	103	0.861845	0.947503	0.902646
7	LJUBLJANA_pleasant_weather	1543	1470	294	3901	73	0.833333	0.952690	0.889023
10	MUNCHENB_pleasant_weather	1192	1090	173	4373	102	0.863025	0.914430	0.887984
6	KASSEL_pleasant_weather	923	830	153	4662	93	0.844354	0.899242	0.870934
8	MAASTRICHT_pleasant_weather	1176	1078	224	4338	98	0.827957	0.916667	0.870056
4	DUSSELDORF_pleasant_weather	1231	1096	206	4301	135	0.841782	0.890333	0.865377
0	BASEL_pleasant_weather	1400	1258	262	4076	142	0.827632	0.898571	0.861644
11	OSLO_pleasant_weather	859	730	134	4745	129	0.844907	0.849825	0.847359
3	DEBILT_pleasant_weather	1101	953	214	4423	148	0.816624	0.865577	0.840388
5	HEATHROW_pleasant_weather	1168	1063	300	4270	105	0.779897	0.910103	0.839984
13	STOCKHOLM_pleasant_weather	972	753	199	4567	219	0.790966	0.774691	0.782744
12	SONNBLICK_pleasant_weather	0	0	0	5738	0	0.000000	0.000000	0.000000
14	VALENTIA_pleasant_weather	276	0	1	5461	276	0.000000	0.000000	0.000000

Comparing CNN & Bayesian Optimization: What Actually Improved

1. Overall performance became more consistent across stations

- After tuning, most stations cluster in a high F1 range (~0.85–0.95).
- Before tuning, performance was more uneven, with several stations well below 0.80.

Interpretation: Bayesian optimization reduced volatility and made the CNN more reliable station-to-station.

2. Biggest gains occurred where the baseline struggled

- VALENTIA: F1 improved from ~0.03 → ~0.28 (+0.25)
- OSLO: 0.72 → 0.83 (+0.11)
- LJUBLJANA, STOCKHOLM, MUNICH: gains of ~0.06–0.07

Interpretation: Optimization primarily fixed underperforming or unstable stations, rather than inflating already-strong ones.

3. High-performing stations improved modestly, not artificially

- MADRID: 0.87 → 0.93 (+0.06)
- BELGRADE: 0.82 → 0.87 (+0.05)
- BASEL: 0.72 → 0.75 (+0.03)

Interpretation: Strong stations stayed strong, with incremental improvements, indicating no overfitting.

4. A few stations saw slight declines but within noise range

- DUSSELDORF, MAASTRICHT, DEBILT: F1 decreased by ~0.01–0.02
- These stations still maintain solid F1 scores (~0.75–0.79)

Interpretation: Minor trade-offs occurred as the model optimized globally, but no station degraded materially.

5. Precision and recall remain well balanced

- Most tuned stations show precision ≥ 0.82 and recall ≥ 0.89
- Indicates the model is not simply predicting “pleasant” too often

Interpretation: Improvements reflect better pattern learning, not threshold gaming.

2.6 Overfitting and Generalization Assessment

Training and validation curves were examined to assess overfitting risk:

- Training and validation binary accuracy remained closely aligned throughout training.
- Validation loss stabilized early, supported by early stopping.
- No evidence of divergence between training and validation performance was observed.

These results suggest that Bayesian optimization improved model generalization rather than simply increasing model complexity.

2.7 Interpretation and Operational Implications

The optimized CNN model demonstrates that deep learning can effectively capture complex temporal and multivariate weather patterns relevant to flight safety decisions.

Key implications for Air Ambulance operations include:

- The ability to generate station-specific probability estimates for safe flying conditions
- Improved reliability compared to manually tuned deep learning models
- Complementarity with Random Forest models: while Random Forests offer transparency and feature interpretability, CNNs provide stronger pattern recognition across time-dependent inputs

Together, the optimized CNN and Random Forest models form a robust decision-support toolkit.

2.8 Key Takeaways from Part 2

- Bayesian hyperparameter optimization substantially improved CNN performance.
- Retaining a multi-label formulation preserved the operational meaning of predictions.
- Optimized deep learning models showed strong generalization without overfitting.

- Deep learning adds value by modeling temporal and multivariate interactions that are difficult to capture with tree-based methods alone.

Part 3: Iteration Strategy and Model Refinement

3.1 Iterative Data Decomposition Strategy

To further improve predictive reliability and operational usefulness, the next phase of model development should focus on **iterative decomposition of the dataset into smaller, more targeted components**. Rather than attempting to optimize a single global model, breaking the data into meaningful subsets allows each model to specialize and reduce noise introduced by heterogeneous conditions. The following decomposition strategies are recommended:

Temporal Segmentation

- **Seasonal subsets** (e.g., winter vs summer conditions)
- **Extreme weather periods** versus typical operating conditions
- **Decadal retraining windows** to account for climate trends

This would allow models to learn condition-specific patterns, particularly for temperature- and precipitation-driven risk.

Spatial Segmentation

- **Regional clusters of stations** with similar climate profiles
- **Station-specific datasets** for high-traffic or critical base locations

The Random Forest results demonstrated that station-specific models with long historical records are highly stable, while multi-station models benefit from capturing regional variability.

Feature Subsetting

- Iterating on **core safety variables only** (e.g., temperature, precipitation, cloud cover)
- Testing reduced feature sets to assess model robustness and interpretability
- Evaluating whether secondary variables add meaningful predictive value

3.2 Model Selection for Iterative Testing

Different models are better suited to different stages of iteration and decision support.

Random Forests should be the **primary tool for early-stage iteration** due to their transparency and stability.

Recommended uses:

- Rapid testing of new feature subsets

- Identifying dominant predictors under different conditions
- Evaluating whether additional variables meaningfully improve performance
- Communicating results to non-technical stakeholders

The part 1 analysis showed that optimized Random Forests surface **clear, operationally intuitive variables**, making them ideal for narrowing the feature space before deeper modeling.

Deep Learning Models (CNNs) should be used **after feature relevance is established**, focusing on temporal dynamics and interactions.

Recommended uses:

- Capturing sequential weather patterns (e.g., temperature trends preceding precipitation)
- Testing lagged or rolling-window features
- Refining probability estimates once key variables are confirmed

The part 2 results demonstrated that CNN performance improves dramatically once hyperparameters are optimized, but the model is most effective when applied to a well-curated feature set rather than exploratory feature discovery.

3.3 Priority Variables for Air Ambulance Flight Safety

Synthesizing insights from both modeling approaches, the following variables should receive **highest operational priority** when determining whether conditions are safe to fly.

Tier 1: Critical No-Fly Indicators

These variables consistently dominated both Random Forest feature importance and CNN predictive performance.

1. **Precipitation**
    - Strongest predictor in station-specific models
    - Directly impacts visibility, rotor performance, and landing safety
    - Recommended as an immediate no-fly trigger when thresholds are exceeded
2. **Maximum Temperature**
    - Affects air density, lift capacity, and engine performance
    - Highly influential across multiple stations in regional models
    - Particularly important during summer and extreme heat events

Tier 2: Conditional Risk Indicators

These variables provide contextual risk assessment rather than absolute go/no-go decisions.

3. **Cloud Cover**
    - Proxy for visibility and broader weather systems
    - Became more prominent after Random Forest optimization
    - Useful for assessing marginal flying conditions
4. **Mean Temperature Stability**

- Indicates atmospheric stability over the day
- Rapid fluctuations may signal changing or deteriorating conditions

Tier 3: Supporting Contextual Indicators

These variables add confirmation value but should not be used in isolation.

5. **Global Radiation**
   - Reflects overall weather system clarity
   - Helpful for confirming stable, high-visibility conditions
6. **Sunshine Duration**
   - Secondary visibility indicator
   - Most useful in conjunction with cloud cover and precipitation data

3.4 Iterative Decision-Support Framework

Based on the above, an iterative modeling framework for Air Ambulance operations is recommended:

- **Stage 1:** Use Random Forests to validate variables and thresholds within specific regions or seasons.
- **Stage 2:** Apply optimized CNNs to model temporal risk patterns using the validated variable set.
- **Stage 3:** Deploy station-specific models for daily operational decisions and regional models for broader readiness planning.
- **Stage 4:** Retrain and re-optimize models periodically as new weather data becomes available and climate patterns evolve.

3.5 Key Takeaways from Iteration Analysis

- Iteration should focus on **problem decomposition**, not just parameter tuning.
- Random Forests are best suited for interpretability and early-stage refinement.
- CNNs add value once the feature space is disciplined and well-understood.
- Precipitation and temperature consistently emerge as the most critical safety variables.
- A tiered, iterative modeling approach provides the strongest support for real-world Air Ambulance flight decisions.